

INGUVA SATYAVENKATA SAI KUMAR

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Experience

NStarX

Jun 2024 – Present

Software Engineer

- Built UI pages and integrated backend APIs for the NYC project, an AI chat interface converting natural language queries into SQL, with Microsoft and Google login integrated for secure authentication and role-based access control.
- Developed UI screens and integrated APIs for GenAI Lab, a Micro Frontend platform in React.js where each AI service has its own independent frontend and Python backend, building and connecting several APIs using FastAPI throughout.
- Integrated ArgoCD into the backend for select APIs to enable one-click deployments, and built an admin control panel that deploys each AI use case based on individual user requests directly within the unified platform interface.
- Integrated AWS S3 for file storage within GenAI Lab, supporting reliable and scalable storage for data generated and used across the different independent AI services running on the shared production platform infrastructure today.
- Developed UI screens and integrated backend APIs for the Digital Human project, building and connecting endpoints that power the real-time monitoring dashboard used by the team to track overall system health across all deployments.
- Integrated pod and cluster metrics and logs using event streaming, and implemented continuous polling to keep dashboard data updated in real time, always ensuring it reflected the most current and accurate system status data.
- Designed the Vue.js dashboard displaying pod and cluster metrics and resource usage across the platform, and also contributed to the UE5 avatar work supporting the voice-driven 3D character used throughout the entire project.

Projects

FilmSense

May 2026 – Present

- Built FilmSense, an AI-powered movie assistant that lets users search by Genre, Character, and Theme together instead of just a title, using semantic matching over actual movie content rather than watch history or basic filters.
- Engineered a hybrid retrieval pipeline using Sentence Transformers for embeddings and a Qdrant vector database, understanding both exact titles and vague descriptions while handling typos and safeguarding against wrong matches.
- Built a Movie Details page pulling and organizing overview, director, writers, cast, ratings, votes, runtime, and trailer automatically, giving users a complete picture of any movie found through the semantic search interface.
- Developed a System Dashboard with live CPU, RAM, disk, and GPU monitoring, since FilmSense runs on self-managed infrastructure that needed real-time visibility into resource usage and overall system health continuously today.
- Built the backend using FastAPI and the frontend using React with TypeScript, designing the retrieval pipeline, handling messy search inputs, and keeping API responses fast across the entire end-to-end system built independently.

Tech Stack: TypeScript, React, Python, FastAPI, Hugging Face Transformers

Certificate

- Introduction to Applied Data Science with Python [🔗](#)

Skills

- **Languages:** JavaScript, TypeScript, Python, HTML, CSS
- **Frameworks:** React, Vue.js, NextJS, Node.js, Flask, FastAPI
- **Tools:** Docker, Postman, GitHub, Unittest, Pytest, VS Code, Jest
- **Platforms:** AWS, Linux, Google Cloud

Education

Gitam University

B.Tech in Electronics and Communication Engineering; CGPA: 8.10

Oct 2020 – Apr 2024

Hyderabad